About This Document

The Pseudocode / Technical section tests data structures, algorithms, complexity and program output - also useful for the technical interview.

Data Structures and Complexity

14 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Time complexity of binary search on a sorted array of n elements:

(A) O(n)
(B) O(log n) ✔
(C) O(n log n)
(D) O(1)

Answer: (B) O(log n)

Authenticity: Frequently Repeated

Q2. Which data structure follows FIFO?

(A) Stack
(B) Queue ✔
(C) Tree
(D) Graph

Answer: (B) Queue

Authenticity: Community Reported

Q3. Which data structure follows LIFO?

(A) Queue
(B) Stack ✔
(C) Linked List
(D) Heap

Answer: (B) Stack

Authenticity: Community Reported

Q4. Worst-case time complexity of bubble sort:

(A) O(n)
(B) O(n log n)
(C) O(n^2) ✔
(D) O(log n)

Answer: (C) O(n^2)

Authenticity: Community Reported

Q5. Which BST traversal yields sorted order?

(A) Preorder
(B) Postorder
(C) Inorder ✔
(D) Level order

Answer: (C) Inorder

Authenticity: Community Reported

Q6. Average-case lookup time in a hash table:

(A) O(1) ✔
(B) O(n)
(C) O(log n)
(D) O(n log n)

Answer: (A) O(1)

Authenticity: Community Reported

Q7. A tree with n nodes has how many edges?

(A) n
(B) n-1 ✔
(C) n+1
(D) 2n

Answer: (B) n-1

Authenticity: Community Reported

Q8. Which is NOT a linear data structure?

(A) Array
(B) Stack
(C) Queue
(D) Tree ✔

Answer: (D) Tree

Authenticity: Community Reported

Q9. The recurrence T(n) = 2T(n/2) + n solves to:

(A) O(n)
(B) O(n log n) ✔
(C) O(n^2)
(D) O(log n)

Answer: (B) O(n log n)

Authenticity: Frequently Repeated

Q10. Maximum nodes at level L of a binary tree (root at level 0):

(A) 2L
(B) L^2
(C) 2^L ✔
(D) 2^L - 1

Answer: (C) 2^L

Authenticity: Community Reported

Q11. Space complexity of standard merge sort:

(A) O(1)
(B) O(log n)
(C) O(n) ✔
(D) O(n^2)

Answer: (C) O(n)

Authenticity: Community Reported

Q12. Which sort is stable with guaranteed O(n log n)?

(A) Quick sort
(B) Merge sort ✔
(C) Selection sort
(D) Heap sort

Answer: (B) Merge sort

Authenticity: Community Reported

Q13. Which sorting algorithm is fastest on average for large random data?

(A) Bubble
(B) Insertion
(C) Quick sort ✔
(D) Selection

Answer: (C) Quick sort

Authenticity: Community Reported

Q14. A stack is best used to implement:

(A) BFS
(B) Function call/recursion handling ✔
(C) Priority scheduling
(D) Round robin

Answer: (B) Function call/recursion handling

Authenticity: Community Reported

Pseudocode Output and Logic

8 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. function f(n){ if(n<=1) return 1; return n*f(n-1); } f(4) returns:

(A) 12
(B) 24 ✔
(C) 16
(D) 6

Answer: (B) 24

Authenticity: Frequently Repeated

Q2. for(i=0; i<3; i++) print(i); Output:

(A) 1 2 3

(B) 0 1 2 ✔

(C) 0 1 2 3

(D) 0 1

Answer: (B) 0 1 2

Authenticity: Community Reported

Q3. x=10; y=20; x=x+y; y=x-y; x=x-y; Final x, y:

(A) x=10,y=20

(B) x=20,y=10 ✔

(C) x=30,y=10

(D) x=20,y=20

Answer: (B) x=20,y=10

Authenticity: Frequently Repeated

Q4. sum=0; for i in 1..n: for j in 1..n: sum++. Final sum:

(A) n

(B) 2n

(C) n^2 ✔

(D) n log n

Answer: (C) n^2

Authenticity: Community Reported

Q5. Postfix of (A + B) * C:

(A) AB+C* ✔

(B) ABC+*

(C) A+BC*

(D) AB*C+

Answer: (A) AB+C*

Authenticity: Community Reported

Q6. count=0; n=18; while(n>0){ count++; n=n/2; } (integer div). count:

(A) 4

(B) 5 ✔

(C) 6

(D) 3

Answer: (B) 5

Authenticity: Frequently Repeated

Q7. a=[1,2,3,4]; s=0; for x in a: s+=x; print(s). Output:

(A) 6

(B) 10 ✔

(C) 24

(D) 4

Answer: (B) 10

Authenticity: Community Reported

Q8. function g(n){ if(n==0) return 0; return n + g(n-1);} g(5):

(A) 10

(B) 15 ✔

(C) 20

(D) 12

Answer: (B) 15

Authenticity: Frequently Repeated

Infosys

Pseudocode & Technical MCQs

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.